

University of Toronto — Cybersecurity Bootcamp

Institution: University of Toronto, School of Continuing Studies

Program: Cybersecurity Boot Camp

Duration: September 2020 – April 2021 (24 Weeks)

Credential: Certificate of Completion

About This Repository

This repository contains all assignments, projects, and deliverables completed during the UofT Cybersecurity Boot Camp — a rigorous 24-week program covering the full spectrum of defensive and offensive security. Work was completed in a hypervisor-based lab environment and submitted via Git.

Skills & Tools Covered

Domain	Tools & Technologies
Operating Systems	Linux (Ubuntu/Kali), Windows Server, Bash, PowerShell
Networking	Wireshark, Cisco Packet Tracer, tcpdump, pfSense
SIEM & Log Analysis	Splunk, ELK Stack (Elasticsearch, Logstash, Kibana)
Penetration Testing	Metasploit, Nmap, Nessus, Burp Suite, John the Ripper
Web Security	OWASP Top 10, SQL Injection, XSS, CSRF, Burp Suite
Cloud & Infrastructure	Azure, Docker, Ansible
Cryptography	OpenSSL, GPG, PKI, hashing algorithms
Forensics & Incident Response	Autopsy, Volatility, Wireshark
GRC	NIST Framework, Risk Assessment, Security Policy

Repository Structure

Week	Topic	Summary
<u>W1</u>	Threat Landscape Research	OSINT, threat actor profiling, attack taxonomy
<u>W2</u>	Assessing Security Culture	Security awareness, policy frameworks, risk culture
<u>W3</u>	A High Stakes Investigation	Digital forensics, log analysis, timeline reconstruction
<u>W4</u>	Linux Systems Administration	User management, permissions, cron jobs, hardening
<u>W5</u>	Archiving and Logging Data	tar, gzip, syslog, log management, auditd
<u>W6</u>	Advanced Bash	Shell scripting, privilege escalation, automation
<u>W7</u>	Windows System Administration	PowerShell, Active Directory, Group Policy, RSAT
<u>W8</u>	Network Fundamentals	OSI model, TCP/IP, Wireshark, subnetting
<u>W9</u>	Network Fundamentals II	Routing, switching, firewalls, VPNs
<u>W10</u>	Cryptography	Symmetric/asymmetric encryption, PKI, GPG, hashing
<u>W11</u>	Network Security	IDS/IPS, firewall rules, network hardening
<u>W13</u>	ELK Stack Project ★	Built a cloud-based SIEM with ELK on Azure
<u>W14</u>	Web Development	HTTP, APIs, microservices, CRUD, curl
<u>W15</u>	Web Vulnerabilities & Hardening	OWASP Top 10, Burp Suite, SQL injection, XSS
<u>W16</u>	Penetration Testing	Nmap, Metasploit, exploitation, reporting
<u>W18</u>	SIEMs and Splunk I	Splunk queries, dashboards, alert creation
<u>W19</u>	SIEMs and Splunk II	Advanced SPL, correlation rules, threat hunting
<u>W20</u>	Red vs. Blue Project ★	Full attack/defense simulation with Kibana analysis
<u>W24</u>	Final Project ★	Penetration test + defensive analysis + executive report

★ = Major project deliverable

Highlighted Projects

● ELK Stack Project (Week 13)

Deployed a cloud-based monitoring and SIEM solution on Microsoft Azure using the ELK stack. Configured Filebeat and Metricbeat to collect logs from multiple VMs, then created Kibana dashboards for threat visibility. [View Project →](#)

● ● Red vs. Blue Project (Week 20)

A full offensive/defensive simulation: attacked a vulnerable target machine using Metasploit and manual exploitation techniques, then switched roles to analyze the attack data in Kibana and propose defensive hardening measures. [View Project →](#)

⚡ Final Project (Week 24)

Conducted a full penetration test against a target environment, documented findings in a professional pentest report, and delivered an executive summary with prioritized remediation recommendations. [View Project →](#)

Contact

Feel free to connect on [LinkedIn](#) or reach out via GitHub if you have questions about any of the work in this repository.